

PAPER CODE: BET-C610
EVEN SEMESTER EXAMINATION (MAY-2026)
CLASS: B.TECH. SEMESTER: VI
PAPER TITLE: EMBEDDED SYSTEMS

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
 (5 x 6 = 30 Marks)

		CO	BL
Q-1	Discuss Operating system Architecture approach. Which approach of OS designing is widely used these days is explained.	CO2	L2
Q-2	A message queue in an RTOS can hold up to 10 messages. If Task X sends 7 messages and Task Y processes 3 messages, how many messages remain in the queue?	CO1	L1
Q-3	Explain why microcontrollers are preferred in embedded systems while microprocessors are used in general-purpose computing.	CO3	L1
Q-4	Differentiate between a simple Finite State Machine (FSM) and an FSMD in terms of functionality and application.	CO4	L2
Q-5	Discuss Addressing modes in 8051 microcontroller.	CO3	L1
Q-6	Develop an 8051 assembly program to generate a square wave on a pin of Port 1 with a fixed time delay.	CO3	L2
Q-7	What characteristics make an operating system suitable for real-time applications?	CO5	L2
Q-8	Discuss the advantages of RISC over CISC in modern embedded systems.	CO6	L1
Q-9	Write an assembly program for 8051 using register addressing mode to perform addition of two numbers	CO5	L2
Q-10	Write an 8051 program to transfer a block of data from one memory location to another	CO5	L1

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = 40 Marks)

		CO	BL
Q-1	In an 8051 system with a 12 MHz clock, Timer 0 is configured in mode 2 (8-bit auto-reload). What is the maximum delay possible?	CO1	L4
Q-2	A push button is connected to pin P1.0 of an 8051 microcontroller. Write a program to check the switch status: (A) If the switch is pressed (0), send 0xFF to Port 2 (B) If the switch is not pressed (1), send 0x00 to Port 2	CO2	L2
Q-3	Explain with a diagram of embedded system development life cycle.	CO3	L4
Q-4	An 8051 microcontroller has 128 bytes of internal RAM and 4KB of ROM. If the microcontroller is connected to an external 8KB of ROM, calculate the total memory space available for the program storage and data storage. How much of this memory is accessible by the CPU at any given time?	CO2	L3
Q-5	Create a program for 8051 to display alternate LED patterns (0xAA and 0x55) on 8 LEDs connected to Port 3.	CO5	L3
Q-6	The 8051 microcontroller has an interrupt system that manages external and internal events. List the five interrupt sources in the 8051 and explain how the interrupt priority can be controlled.	CO7	L3
Q-7	An embedded system has an NRE cost of ₹2,00,000 and a recurring cost of ₹200 per unit. If 5,000 units are produced, calculate the total cost per unit.	CO9	L2
Q-8	Write a program to find the LCM of two numbers. How can you optimize the solution using the GCD?	CO10	L3